

Adaptation of StyleGAN3 for Processing Multidimensional Remote Sensing Images

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Abstract—Deep learning models for high-resolution remote sensing classification frequently face challenges related to data scarcity and severe class imbalance. This work proposes a unified StyleGAN3 architecture adapted for multispectral image synthesis and classification to address these limitations. The method integrates a dense classification head into the discriminator, transforming it into a dual-purpose architecture, and implements a class-balanced adversarial training strategy where the generator is conditioned to sample from a uniform class distribution.

To evaluate the system, a domain-specific framework assesses FID, Precision, and Recall using a custom multispectral judge model. Experimental results on eight river basin datasets demonstrate that the proposed method achieves a mean Average Accuracy (AA) of 90.4%, significantly outperforming standard baselines (ResNet, ViT) and previous GAN-based frameworks (ResBaGAN, EffBaGAN) in data-scarce scenarios.

Index Terms—Multispectral Remote Sensing, StyleGAN3, Class Imbalance, Data Augmentation, Generative Adversarial Networks (GANs), Deep Learning Classification

I. INTRODUCTION

Remote sensing is a fundamental technique for gathering information about objects or geographical areas without physical contact. By utilizing sensors mounted on satellites, aircraft, or drones, researchers can collect high-resolution imagery of the Earth's surface. This technology is essential for global monitoring tasks, such as tracking urban development [1], monitoring agricultural changes [2], and conducting environmental studies [3]. Within this field, multispectral sensors play a crucial role. Unlike conventional cameras that are limited to Red, Green, and Blue (RGB) channels, multispectral sensors capture data across various wavelengths, including the infrared spectrum. These Multispectral Images (MSI) provide a rich dataset that allows for a more detailed and accurate analysis of the terrain compared to standard optical photography.

In the context of remote sensing, classification involves assigning a specific label to every pixel in an image to identify its land-cover type. For instance, in a river basin study, the goal is to distinguish between various elements such as different tree species, water bodies, soil paths, or artificial structures. Historically, classification relied on classical mathematical and statistical methods. However, in recent years, Deep Learning (DL) has emerged as the dominant approach, particularly through the use of Convolutional Neural Networks (CNNs). These models are highly effective because they can automatically learn complex spatial and spectral patterns without the need for manual feature engineering.

Despite these advancements, challenges persist regarding data availability, particularly in drone-captured imagery which is frequently characterized by a scarcity of labeled ground truth and significant class imbalance [4], [5]. These constraints pose substantial challenges for deep learning frameworks; consequently, classifiers frequently exhibit bias, resulting in disparate precision metrics across different categories [6]–[8]. Furthermore, the use of high spatial resolution imagery induces greater intraclass variability—where entities within the same category appear diverse—and interclass similarity, where distinct classes display overlapping spectral signatures [4]. Despite these obstacles, deep learning architectures generally demonstrate superior generalization capabilities compared to traditional machine learning techniques in these complex environments [9].

To address these challenges, a variety of data augmentation techniques have been proposed to artificially increase the dataset size. A fundamental distinction exists between classical augmentations techniques and deep learning-based approaches. Classical techniques encompass geometric transformations, such as rotation and flipping [10], as well as statistical techniques based on data imputation [11]. In contrast, deep learning-based augmentation relies on constructing models capable of synthesizing entirely new samples. Prominent approaches include Variational Autoencoders (VAEs), Generative Adversarial Networks (GANs), and Diffusion Models [8].

Recently, GAN-based augmentation has gained popularity due to its ability to generate high-quality synthetic data. A GAN comprises two competing networks: a generator and a discriminator. These systems can be designed with two different objectives: refining the generator to produce synthetic samples that expand the dataset effectively [12]–[14], or extending the discriminator to act as a multi-class classifier that leverages adversarial training to improve generalization [6], [7]. However, these classification-centric approaches often neglect the qualitative assessment of the generator. This omission is critical, as the production of high-fidelity, diverse samples is essential to challenge the discriminator effectively, forcing it to capture complex semantic dependencies that improve its generalization.

Similar to GANs, diffusion models have recently established themselves as robust tools for data augmentation [8], [15]. Functioning as generative networks, they synthesize samples to enhance downstream classification tasks. While they share structural similarities with autoencoder decoders, diffusion

models are uniquely defined by their ability to reconstruct high-likelihood images from pure noise through the iterative reversal of a perturbation process.

However, diffusion models are inherently slower to train because they model the full data distribution by reversing a progressive degradation process. Unlike architectures that seek a direct mapping, these models must learn to estimate and remove noise at thousands of distinct intensity levels [16]. This objective, learning data density gradients across the entire latent space rather than simply generating plausible images, demands substantially greater computational time and a larger number of training iterations to achieve convergence [17].

In contrast to the heavy computational demands of diffusion models, adversarial architectures remain superior in terms of efficiency. The StyleGAN family [18]–[20], in particular, has set the benchmark for high-fidelity image synthesis with significantly lower computational overhead. StyleGAN3 represents a major evolution by eliminating “texture sticking”, an artifact where generated details appear glued to the screen coordinates rather than moving naturally with the object. This architecture combines these visual improvements with the inherent efficiency of GANs, offering much faster training convergence and inference speeds compared to diffusion-based approaches.

The evaluation of generative models, such as GANs and diffusion probabilistic models, typically relies on quantitative metrics that estimate the divergence between real and synthetic data distributions, most notably the Fréchet Inception Distance (FID) [21], Inception Score (IS) [22], and Kernel Inception Distance (KID) [23]. These metrics avoid direct pixel-space comparison by employing a judge model as a feature extractor to map images into a high-dimensional latent space. The standard implementation utilizes the Inception-v3 architecture, pretrained on the ImageNet dataset, extracting representations from deep pooling layers to compute the statistical distance between the generated and reference datasets [21].

However, the reliance on an ImageNet-pretrained Inception-v3 judge presents critical limitations for multispectral remote sensing contexts. Firstly, there is a dimensionality mismatch: the standard judge is trained exclusively on 3-channel RGB data, making it architecturally incompatible with multispectral tensors containing additional bands without invalidating the pretrained weights. Secondly, a significant domain shift exists between the object-centric, horizontal-view photography of ImageNet and the texture-centric, nadir-view patterns of UAV-acquired multispectral imagery. Consequently, the features extracted by a standard RGB judge fail to encapsulate the spectral complexity and distinct biophysical patterns inherent to remote sensing data, necessitating domain-specific evaluation strategies.

In this article, a modification of the StyleGAN3 architecture [20] to classify multispectral imagery captured from UAVs is proposed. The objective is to leverage the generative capabilities of StyleGAN3 to produce high-quality, diverse synthetic images, thereby addressing the challenges of class imbalance and data scarcity without the need to train a separate generative

model and classifier.

To rigorously evaluate the generative performance, we incorporate FID alongside Precision and Recall metrics into the training and validation loops. Specifically, we adopt the improved formulation of Precision and Recall by [24] to explicitly disentangle visual fidelity from distributional diversity. Following the method established in [8], the standard evaluation pipeline is adapted by employing a specialized judge model pretrained on multispectral data. This ensures that the extracted features capture the specific spectral-spatial semantics of the UAV domain, while simultaneously addressing class imbalance during the assessment process.

A. Contributions

This work proposes a unified StyleGAN3 architecture for high-resolution multispectral image synthesis and classification, specifically designed to mitigate the challenges of data scarcity and class imbalance in remote sensing contexts. The key contributions of this work can be summarized as follows:

- *StyleGAN3 Discriminator Adaptation for Multispectral Classification:* We propose a modification of the StyleGAN3 architecture by integrating a dense classification head into the discriminator. This adaptation extends the model’s capability to process multispectral imagery, transforming the framework into a dual-purpose architecture that simultaneously learns high-fidelity image synthesis and accurate classification without requiring external models.
- *Class-Balanced adversarial Training:* We introduce a class-balanced training strategy where the generator is constrained to sample from a uniform class distribution rather than the inherent long-tailed distribution of the dataset. By exposing the discriminator (and its classifier component) to a balanced stream of synthetic examples during optimization, this approach actively mitigates class imbalance bias, ensuring robust learning representation across all classes, including underrepresented ones.
- *Domain-Specific Quality-Performance Analysis:* We employ a framework to assess FID, Precision, and Recall using a custom multispectral judge. Beyond standard quality monitoring, this study systematically investigates the correlation between these generative metrics and classification accuracy. This approach aims to identify the specific balance between visual fidelity and sample diversity required to maximize the effectiveness of synthetic data augmentation in imbalanced scenarios.

II. RELATED WORK

This section outlines the related work establishing the context for our approach. We begin by reviewing generative frameworks designed for handling data imbalance in multispectral remote sensing, covering both GAN-based methods and diffusion models. Subsequently, we describe the advanced generative architecture that underpins our specific implementation.

A. ResBaGAN

The ResBaGAN architecture [6] is a generative framework designed for multispectral image classification. Following the Balancing GAN (BAGAN) framework [25], ResBaGAN employs an autoencoder-based initialization strategy. Unlike standard GANs that initialize weights randomly—often leading to mode collapse in minority classes—this pretraining phase allows the model to learn a meaningful feature manifold covering all categories prior to adversarial training, thereby ensuring that underrepresented classes are not ignored during the optimization process.

Once initialized, the architecture incorporates classification capabilities directly into the discriminator through two distinct architectural topologies. The first approach follows the standard Auxiliary Classifier GAN (AC-GAN) scheme [26], featuring a dual-output mechanism where adversarial validity and class probabilities are decoupled. In contrast, the second strategy adopts the unified formulation proposed in the original BAGAN framework [25]. This approach consolidates the objectives into a single output layer, effectively treating the “fake” category as just an additional class within the decision boundary. By combining this initialization strategy with intrinsic data augmentation, ResBaGAN demonstrated that generative classifiers could achieve robust performance even in scenarios characterized by severe spectral similarity and data scarcity.

B. EffBaGAN

EffBaGAN [7] emerges as a direct evolution of the ResBaGAN architecture, specifically engineered to address the computational overhead associated with deep generative models in remote sensing. While retaining the core balancing and classification strategies of its predecessor, the primary objective of EffBaGAN is to optimize resource utilization, significantly reducing computational costs and parameter count while maintaining competitive classification performance.

To achieve this efficiency, the framework fundamentally restructures the adversarial backbones. The discriminator is implemented as a Reduced EfficientNet Discriminator, which adapts the lightweight EfficientNet-B0 topology to process multispectral imagery. Complementing this, the generator is redefined as a Residual EfficientNet Generator, which utilizes a residual path to enhance the quality of synthetic samples.

C. EMViT-DDPM

The EMViT-DDPM framework [8] is a diffusion-based methodology designed to address the challenges of data scarcity and class imbalance in land cover classification using multispectral imagery. The methodology is divided into two main stages. First, a diffusion-based generative model, named MVIT-DDPM (Multispectral Vision Transformer Denoising Diffusion Probabilistic Model), is trained using a Vision Transformer architecture to synthesize high-quality and perceptually realistic multispectral images.

Subsequently, this model is used to augment the original dataset, enabling the training of classifiers in a scenario where

bias towards majority classes is mitigated through the inclusion of synthetic samples. To control this process, the authors introduce the Equilibrium-Based Data Augmentation (EBDA) algorithm, which determines the exact quantity of synthetic samples to generate for each class based on a proportion parameter p and the size of the majority class (n_{max}).

Once a pool of synthetic samples is generated, the algorithm calculates the number of new samples (n_{synth}^i) required for a class i using the following formula:

$$n_{synth}^i = \lfloor p \cdot n_{max} - |C_i| \rfloor, \quad (1)$$

where $|C_i|$ represents the number of existing real samples in that class. The parameter p allows adjusting the desired equilibrium level:

- For a value of $p = 0.5$, minority classes are augmented to reach, at a minimum, half the number of samples of the original majority class, leaving the majority class unchanged.
- For a value of $p = 2$, the training target is set to double the size of the original majority class; this implies that all classes (including the majority one) receive synthetic samples until reaching this new threshold ($2 \cdot n_{max}$).

D. StyleGAN3

While diffusion models like EMViT-DDPM set a new standard for synthesis quality, and frameworks like ResBaGAN and EffBaGAN offer specialized mechanisms for class balancing, our proposal seeks to leverage the architectural advancements of modern Generative Adversarial Networks. Specifically, we adopt StyleGAN3 [20] as the backbone of our system. This architecture is distinguished by its alias-free design, which enables strictly equivariant generation. By treating images as continuous signals, StyleGAN3 ensures that high-frequency details—such as road markings or vegetation textures—rotate and translate coherently with the object, rather than appearing fixed to the pixel grid. In the context of remote sensing, this property ensures that the shape and texture patterns of the original images are faithfully reflected in the synthetic samples, maintaining their structural fidelity even under geometric transformations.

The architecture of the StyleGAN3 generator is illustrated in Figure 1. A defining feature of the StyleGAN family is the style-based mechanism, wherein an intermediate latent vector w modulates the convolution weights. This approach allows for the disentanglement of semantic features. In the context of remote sensing, if w encodes a specific land cover attribute, such as “dense urban”, the modulation amplifies the filters associated with geometric structures while suppressing signals related to vegetation.

Unlike its predecessors (StyleGAN [18] and StyleGAN2 [19]), StyleGAN3 addresses the “texture sticking” artifact. In previous models, high-frequency details—such as tree canopy patterns or lines on the asphalt—often appeared fixed to pixel coordinates rather than moving coherently with the object’s deformation. To solve this, StyleGAN3 treats the image as a continuous signal, enforcing strict equivariance: a geometric

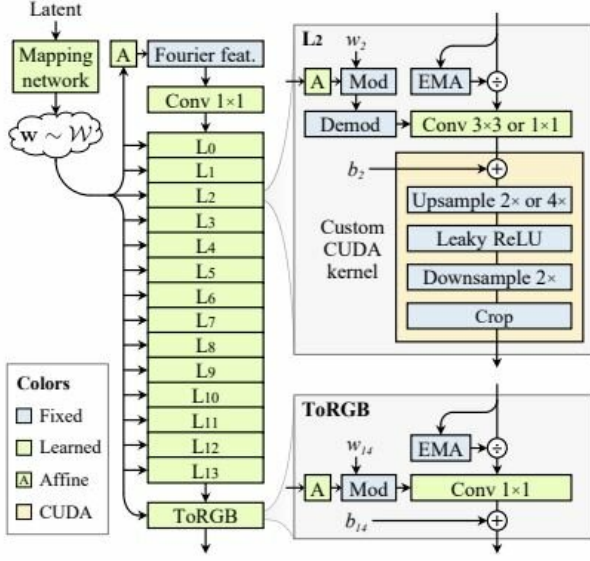


Fig. 1. Diagram of the StyleGAN3 [20] generator architecture, showing the signal flow and the operations within each block.

transformation in the input (translation or rotation) results in an equivalent transformation in the output.

To formalize this constraint, the synthesis network is defined as a continuous operation g parameterized by style weights w . If t represents a geometric transformation (e.g., translation or rotation) applied to the continuous input domain z , the network is considered equivariant if the transformation of the input yields a commensurate transformation of the generated output:

$$g(t[z]; w) = t[g(z; w)].$$

This formulation implies that the signal processing must be strictly alias-free to satisfy the Nyquist-Shannon [27] sampling theorem. By eliminating aliasing, StyleGAN3 ensures that geometric manipulations in the latent representation propagate faithfully to the generated imagery, preventing the emergence of spurious high-frequency artifacts.

To implement this framework, the generator employs Fourier features as input, which are interpreted as frequencies. Instead of optimizing a learned constant tensor, these features are sampled at initialization and remain fixed throughout the training process. Furthermore, the architecture removes specific sources of aliasing, such as per-pixel noise injection, and utilizes normalization layers preceding convolution operations (as shown in Figure 1).

Finally, the generator comprises synthesis layers L_0 to L_{12} followed by a ToRGB block. The signal flow is controlled by a flexible filtering schedule defined by cutoff frequencies (f_c), which intuitively determine the level of detail allowed at each stage. Initially, early layers use a low cutoff frequency to effectively “block” high-frequency signals, allowing the network to process only low frequencies associated with the global structure of the image (e.g., general terrain shapes). Subsequently, deeper layers progressively increase the cutoff frequency; as the signal passes through the network, these

layers “let through” higher frequencies, reintroducing the necessary information to generate fine details like road markings or individual trees.

Regarding the discriminator, we employ a residual network (ResNet) architecture. This design is motivated by the functional similarity between GAN discriminators and standard image classifiers, where residual connections are known to be beneficial [19]. Unlike the generator’s focus on continuous signal processing, the discriminator utilizes a standard dual-path residual block structure with bilinear downsampling to progressively reduce resolution while extracting hierarchical features.

III. METHODOLOGY

A. Sample Extraction Via Superpixel Segmentation

To prepare the data for the StyleGAN3 architecture, a sample extraction strategy based on superpixel segmentation is employed, following the Waterpixels segmentation workflow detailed in [3]. Superpixels consist of homogeneous pixel groupings that adapt to local image structures [28]. In remote sensing contexts, this approach is frequently preferred over sliding-window extraction [6]–[8], as it mitigates data redundancy and ensures sample purity—most pixels within a superpixel likely belong to a single class—thereby reducing label noise.

The complete extraction and classification workflow is detailed in Figure 2. Initially, the multispectral image is segmented into superpixels. Subsequently, one patch of fixed size (32×32 pixels) is extracted centered on the centroid of each resulting region, meaning that each superpixel constitutes a single training sample. The class label for each patch is determined by the ground truth of the central pixel.

Post-classification, the results are mapped back to the full-resolution image space. Given that the extracted patch serves as a representative sample of the homogeneous region, the predicted class is assigned to all pixels contained within the boundaries of the corresponding superpixel. This mechanism effectively extends the patch-level inference to a dense, pixel-level classification map.

B. Multispectral StyleGAN3 Architecture

To enable image classification within StyleGAN3, an auxiliary classifier is introduced, adhering to the strategy proposed by AC-GAN [26]. This architecture modifies the discriminator to employ dual output heads: one dedicated to adversarial discrimination (real vs. fake) and another for semantic label prediction. This auxiliary classifier comprises a dense layer appended to the discriminator’s output, projecting the extracted latent representations onto the class logits corresponding to the dataset’s semantic categories. This modification transforms the discriminator into a dual-purpose architecture, capable of distinguishing between real and generated images while simultaneously learning to classify high-resolution multispectral images captured by UAVs.

Consequently, the discriminator’s loss function is adapted to explicitly incorporate this classification objective. We modify

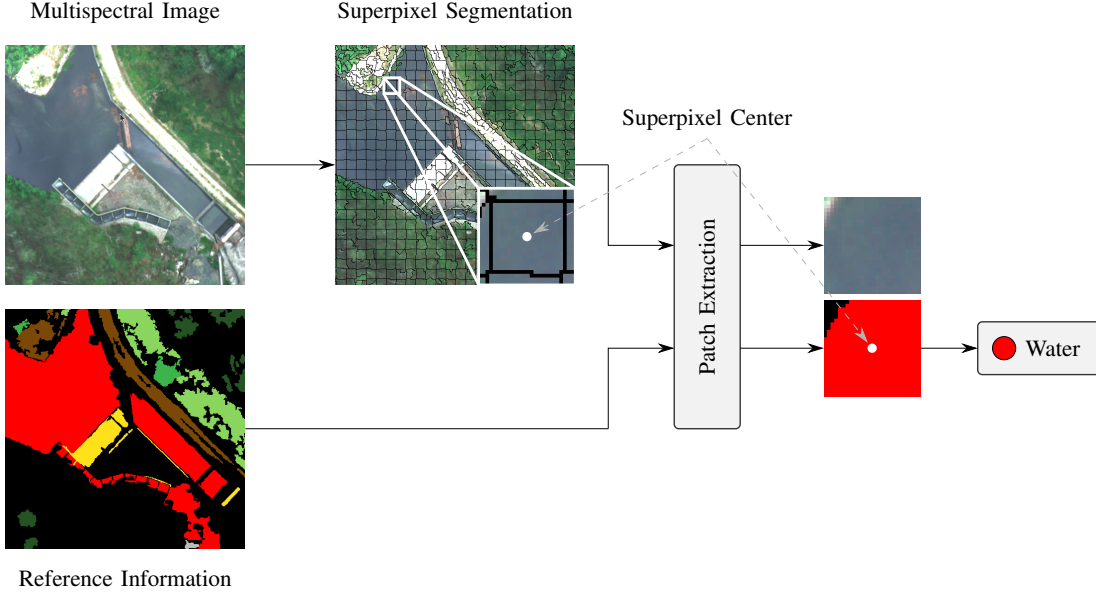


Fig. 2. Sample extraction workflow based on superpixel segmentation: homogeneous regions are first identified within a multispectral image and its reference information. Subsequently, patches are extracted around the computed superpixel centers. The label for each patch is determined by the class of the central pixel as defined in the reference information.

the standard formulation by augmenting the adversarial loss with a weighted categorical cross-entropy term. This combined objective is expressed as:

$$L_D = L_{adv} + \lambda_{cls} L_{cls}.$$

Here, L_{adv} represents the adversarial loss component, while L_{cls} denotes the classification loss. The hyperparameter λ_{cls} is introduced to regulate the weight of the classification penalty within the overall objective. The adversarial term, L_{adv} , retains its standard non-saturating logistic definition. For a real image x , it is computed as:

$$L_{adv}^{real} = \ln(1 + e^{-D(x)}),$$

where $D(x)$ is the discriminator's adversarial output. Similarly, the adversarial component for generated images is defined as $L_{adv}^{fake} = \ln(1 + e^{D(x)})$.

The classification loss term, L_{cls} , is formulated as the categorical cross-entropy over the auxiliary classifier's predictions. For a dataset with N classes, this loss is calculated as:

$$L_{cls} = \sum_{i=1}^N y_i \ln \left(\frac{e^{z_i}}{\sum_{j=1}^N e^{z_j}} \right), \quad (2)$$

where y_i denotes the true class label and z_i represents the logit corresponding to class i produced by the classifier.

Crucially, for both real and generated images, this classification term is weighted by the parameter λ_{cls} to adjust its relative influence on the total loss. This weighting strategy allows the discriminator to balance the dual tasks of distinguishing image authenticity and classifying high-resolution multispectral UAV data. The optimal value for λ_{cls} is determined experimentally to achieve an effective equilibrium between these two objectives.

C. Class-Balanced Adversarial Training

To mitigate class imbalance during training, a balancing strategy is implemented by modifying the generator's input. Specifically, the generator is conditioned to synthesize samples drawn from a uniform class distribution, independent of the real dataset's class frequencies.

It is important to note that we do not aim to balance the combined batch (real plus synthetic) by generating samples inversely proportional to the real data. Such a strategy would force the generator to over-sample minority classes, transferring the class bias to the generative process.

Figure 3 illustrates the training paradigm, highlighting the uniform class prior for generated images in contrast to the original class distribution of real images. By enforcing this uniformity, the generator aids the discriminator in learning robust features across all classes, including those that are underrepresented in the dataset.

D. Quality Metrics for Multispectral Synthetic Samples

Evaluating the quality of synthetically generated images is a critical task in computer vision. Since traditional pixel-level metrics, such as Mean Squared Error (MSE), often fail to capture the perceptual nuances of visual data, we adopt a feature-based evaluation strategy. Following the framework proposed in [8], we adopt a feature-based strategy using FID, Precision, and Recall. This requires a domain-specific "judge" model rather than the standard pretrained Inception-v3. Consequently, we employ a ResNet architecture modified to process 5-band multispectral inputs and trained on our real dataset to extract meaningful feature embeddings.

Regarding the metrics, we compute the Fréchet Inception Distance (FID) to measure the overall statistical alignment

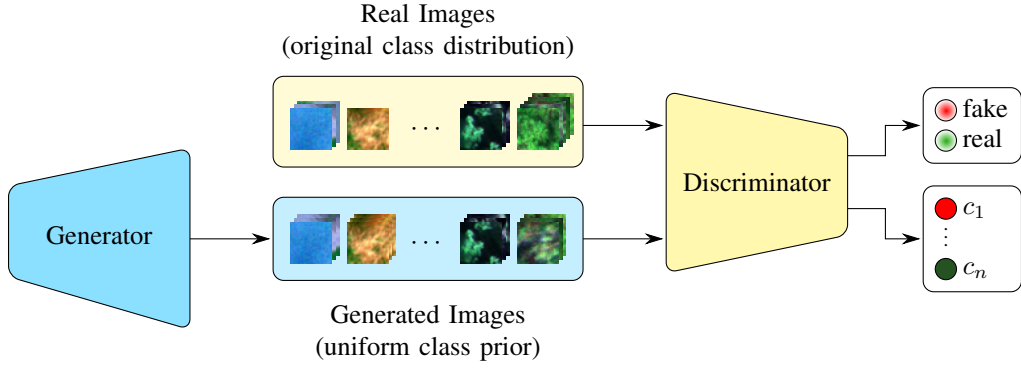


Fig. 3. Class-balanced training strategy: the generator samples latent vectors conditioned on a uniform class distribution, mitigating class imbalance bias during training.

between distributions. To disentangle fidelity from diversity, we also utilize the improved Precision and Recall [24]. While Precision estimates realism (fidelity) and Recall assesses coverage (diversity), their distinct analysis offers a diagnostic advantage over FID. Crucially, this separation allows us to isolate specific failure modes—such as mode collapse (low recall) or the generation of poor-quality artifacts (low precision)—that might otherwise be obscured when relying solely on the aggregated FID score.

IV. EXPERIMENTS

This section details the experimental framework established to validate the proposed StyleGAN3-based classification method. It begins by describing the high-resolution multispectral datasets utilized in the study, followed by a definition of the evaluation metrics for both pixel-level classification and generative quality. Subsequently, the strategy for hyperparameter optimization and class balancing is presented. The section concludes with specifications of the computational infrastructure employed for training and inference.

A. Datasets

The experimental evaluation utilizes high-resolution multispectral imagery originally detailed in [3], captured via Unmanned Aerial Vehicles (UAVs) over river basins in Galicia. A visual overview of the eight scenes comprising the dataset is presented in Fig. 4. These images have served as a benchmark in recent studies addressing class imbalance and data scarcity in multispectral remote sensing classification [6]–[8].

Data acquisition was performed using a Micasense RedEdge-MX sensor, which records information across five spectral bands: Blue (475 nm), Green (560 nm), Red (668 nm), Red Edge (717 nm), and Near-Infrared (840 nm). The imagery provides a spatial resolution of 10 cm per pixel, enabling detailed characterization of land cover features.

The scenes encompass diverse categories, including water bodies, vegetation, rocky surfaces, and human-made structures. A defining characteristic of these datasets is severe class imbalance, where specific classes are significantly under-represented. This distribution poses a challenge for standard

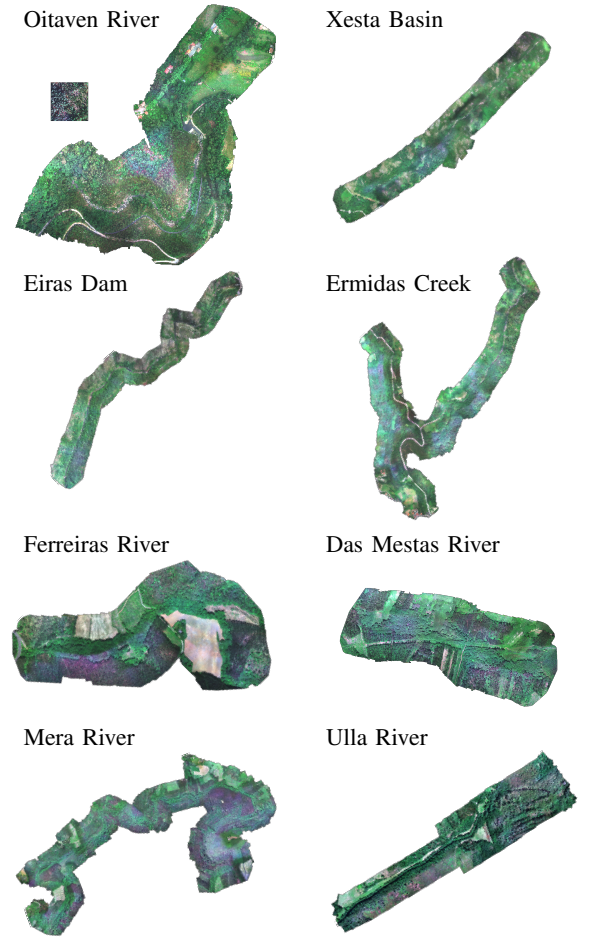


Fig. 4. RGB composite visualizations of the eight river basin scenes used in this study. Each scene is labeled with its corresponding location identifier and is treated as an independent dataset for evaluation.

algorithms, often inducing bias toward majority classes. The specific class distribution for each dataset is summarized in Table I.

Consistent with the sample extraction method described in Section III-A, the datasets were processed to generate superpixel-based patches. To ensure comparability with prior works [6]–[8], the same data partitioning protocols were adopted: 15% of the samples were allocated for training and 5% for validation. This configuration preserves the original class imbalance within each split and simulates a data scarcity scenario.

B. Metrics

StyleGAN3’s performance is evaluated on two fronts: classification accuracy and generative quality.

For classification, pixel-level predictions are compared against ground truth references using two standard remote sensing metrics [29]: Overall Accuracy (OA) and Average Accuracy (AA). To ensure a fair evaluation, the specific pixels used for training (superpixel centroids) are excluded from these calculations.

Generative quality is assessed using Fréchet Inception Distance (FID), Precision, and Recall, as outlined in Section III-D. For the feature extraction stage (judge model), we employ the specific 5-band Modified ResNet pretrained on the Oitavén River dataset, adopting the model weights directly from [8].

Following the protocol in [8], the evaluation involves randomly selecting 200 samples per class from both the real and synthetic pools. These subsets are compared using $k = 3$ for the k-Nearest Neighbors estimation in Precision and Recall. All experiments are repeated five times to account for variability.

C. Hyperparameter Optimization and Balancing Strategy

First, an optimization study was conducted to determine the optimal λ_{cls} weight for the multispectral classification task. A range of values $\{0.01, 0.05, 0.1, 0.5, 1.0\}$ was tested, evaluating the performance based on the auxiliary classifier’s pixel-level accuracy.

Subsequently, experiments were performed to assess the impact of augmenting the auxiliary classifier’s training with synthetic images, examining scenarios both with and without the balancing strategy detailed in Section III-C. Beyond standard performance monitoring, this analysis aims to decouple generative quality from classification utility. By evaluating FID, Precision, and Recall alongside pixel accuracy, the study investigates whether classification improvements depend on high visual fidelity (Precision) or if the classifier benefits more from broader manifold coverage and diversity (Recall), particularly for mitigating data scarcity in underrepresented classes.

Regarding the training protocol, a standard duration of 400 epochs is established for most datasets. The Ulla River dataset serves as an exception, where training is extended to 800 epochs to accommodate its higher scene complexity and

reduced sample size. Model selection is performed by monitoring the Average Accuracy (AA) on the validation set, and the final performance is reported based on the corresponding OA and AA computed on the independent test set.

To ensure robustness and mitigate the influence of training variability, all experiments described above were repeated across five distinct dataset splits. Consequently, results are reported as the mean and standard deviation of these executions.

D. Hardware Configuration

The experiments were conducted utilizing high-performance computing resources provided by the Supercomputing Center of Galicia (CESGA) and a local research cluster. The infrastructure comprised nodes equipped with NVIDIA A100 GPUs (40 GB VRAM) and NVIDIA Tesla V100 GPUs (32 GB VRAM).

Specifically, the local cluster consists of one node with dual NVIDIA A100 GPUs and two nodes with dual NVIDIA Tesla V100 GPUs. To ensure consistency in the reported computational costs, all training times presented in this study refer exclusively to experiments executed using a configuration of two NVIDIA A100 GPUs (40 GB).

V. RESULTS

This section presents a comprehensive evaluation of the proposed framework. The experimental analysis is structured into three main stages to isolate the contribution of each component: first, we determine the optimal contribution of the auxiliary classification loss; second, we conduct an ablation study to assess the impact of the proposed class-balancing strategy on both classification performance and generative quality; and finally, we benchmark the optimized method against state-of-the-art architectures, comparing performance in terms of both classification and image generation metrics.

A. Optimization of Classification Weight

We begin by optimizing the weight of the auxiliary classification loss (λ_{cls}) for the multispectral classification task. Table II summarizes the quantitative results obtained on the test set, reported in terms of Overall Accuracy (OA) and Average Accuracy (AA). The values presented correspond to the mean performance averaged across all datasets. Consequently, the reported standard deviation reflects the variation across all individual experiments, serving as an indicator of the inherent heterogeneity and varying difficulty levels among the different basins.

Globally, it is observed that lower magnitudes of λ_{cls} (specifically 0.01 and 0.05) consistently yield inferior performance metrics. This progression suggests that assigning sufficient weight to the classification loss is critical for the model to learn class-specific representations effectively.

Analyzing the metrics individually reveals distinct behaviors. OA exhibits a consistent, monotonic upward trend as λ_{cls} increases, reaching its peak at $\lambda_{cls} = 1.0$. In contrast, AA improves significantly when increasing λ_{cls} from 0.01 to 0.1, where it reaches its maximum value (85.3%). Beyond

TABLE I
CLASS DISTRIBUTION (NUMBER OF PATCHES/SUPERPIXELS) FOR EACH DATASET

#	Color	Class	Oitavén River	Xesta Basin	Eiras Dam	Ermidas Creek	Ferreiras River	Das Mestas River	Mera River	Ulla River
1		Water	912	210	1 929	447	0	0	0	0
2		Bare soil	484	182	424	472	18 404	4 583	9 856	210
3		Rocks	349	2 170	726	726	0	0	0	0
4		Asphalt	128	467	233	2 418	510	0	241	14
5		Concrete	458	0	111	95	0	0	5	0
6		Tiles	294	0	29	453	0	0	8	0
7		Meadows	6 896	42 821	2 989	10 413	14 087	17 099	22 777	243
8		Native trees	7 573	15 543	8 604	3 154	2 059	1 414	5 129	913
9		Pine	955	0	351	800	11	0	0	81
10		Eucalyptus	3 599	0	39	4 189	19 494	16 133	21 971	0
Total samples:			21 648	61 393	15 435	23 188	54 565	39 299	59 987	1 461

this threshold ($\lambda_{cls} > 0.1$), the AA saturates or exhibits a slight decline, indicating that an excessively high classification weight might favor the majority classes at the expense of the minority classes represented in the AA. Prioritizing the AA metric to ensure robustness against class imbalance, we adopt $\lambda_{cls} = 0.1$ for the subsequent experiments.

TABLE II
IMPACT OF CLASSIFICATION LOSS WEIGHT (λ_{cls}) ON MODEL PERFORMANCE

λ_{cls}	0.01	0.05	0.1	0.5	1.0
OA %	90.8 $\pm$ 3.1	91.9 $\pm$ 3.7	93.3 $\pm$ 3.9	93.6 $\pm$ 4.0	93.8 $\pm$ 4.5
AA %	82.7 $\pm$ 6.9	83.2 $\pm$ 7.5	85.3 $\pm$ 8.0	84.9 $\pm$ 8.4	84.6 $\pm$ 9.5

OA: Overall Accuracy; AA: Average Accuracy.
Values: mean $\pm$ std. dev. (5 runs, 8 datasets).

B. Impact of Balancing Strategy and Evaluation of Synthetic Samples

Building upon the optimized weight configuration, this subsection presents an ablation study to isolate the impact of two critical hyperparameters: the inclusion of synthetic samples in the auxiliary classifier’s training loop and the distribution of those samples (class-balancing). Consequently, we evaluated the performance of the StyleGAN3-based classifier under varying configurations relative to the baseline, labeled as *Original Dist. / Real Only* in Table III, corresponding to the case for which the auxiliary classifier is trained solely on real images using the original class distribution.

Table III reveals a distinct performance hierarchy. First, we analyze the effect of straightforward data augmentation using synthetic samples without altering the class distribution, represented by the *Original Dist. / Real + Syn.* column. In this setup, the generator produces batches following the original dataset priors, and the auxiliary classifier is trained on both real and synthetic images. The results indicate that this strategy consistently outperforms the baseline. While the AA sees a modest increase, the OA improves across the board, achieving

the highest global average of 94.05%. This suggests that in datasets with limited availability—such as the ones presented in this study—injecting synthetic samples acts as a robust regularization technique. It enriches the training manifold and improves the model’s generalization capability on majority classes, effectively serving as a data augmentation mechanism rather than a balancing one.

However, the most significant gains in robustness are observed when the sampling strategy is altered to a uniform distribution, as shown in the *Balanced Dist. / Real + Syn.* column. By forcing the generator to produce a balanced batch of synthetic images for the auxiliary classifier—as explained in Section III-C—we observe a substantial recovery in metrics for minority classes. This is particularly evident in the most severely imbalanced datasets, such as Ferreiras River, Eiras Dam, and Mera River. In these scenarios, where the disparity between majority and minority classes is drastic, the AA improves by approximately 6.8%, 6.1%, and 16.1%, respectively, compared to the baseline. On average, this configuration boosts AA by over 5.11%, confirming that explicitly training the auxiliary classifier on balanced synthetic data is the optimal strategy for ensuring equitable performance across all semantic classes.

Finally, the results for the *Balanced Dist. / Real Only* configuration highlight the interaction between the generator and the discriminator when synthetic samples are not explicitly used for classification. Although the generator produces balanced samples, the auxiliary classifier is not actively supervised by them. Consequently, the shared feature extraction layers of the discriminator do not benefit from this balanced exposure. This leads to sub-optimal performance, further emphasizing that the benefits of the proposed method stem from the active supervision of synthetic samples rather than passive feature alignment.

Beyond classification metrics, it is crucial to analyze how these strategies affect the structure of the learned manifold. Table IV summarizes the FID, Precision, and Recall metrics obtained for each configuration. Several distinct trends emerge

TABLE III
IMPACT OF CLASS BALANCING AND SYNTHETIC DATA INTEGRATION ON CLASSIFICATION PERFORMANCE

Dataset		Original Dist. Real Only	Balanced Dist. Real Only	Original Dist. Real + Syn.	Balanced Dist. Real + Syn.
Eiras Dam	OA %	94.4 $\pm$ 0.8	95.0 $\pm$ 0.3	95.2 $\pm$ 0.6	93.9 $\pm$ 1.2
	AA %	84.0 $\pm$ 1.2	86.5 $\pm$ 1.4	85.9 $\pm$ 1.8	90.0 $\pm$ 1.8
Ermidas Creek	OA %	96.7 $\pm$ 0.3	97.0 $\pm$ 0.2	97.4 $\pm$ 0.4	96.8 $\pm$ 0.6
	AA %	93.3 $\pm$ 0.8	94.0 $\pm$ 0.2	93.2 $\pm$ 0.8	96.1 $\pm$ 0.4
Ferreiras River	OA %	89.5 $\pm$ 0.4	89.8 $\pm$ 0.6	91.1 $\pm$ 0.2	89.5 $\pm$ 1.3
	AA %	77.1 $\pm$ 0.8	64.1 $\pm$ 0.6	78.0 $\pm$ 0.9	83.9 $\pm$ 5.0
Mera River	OA %	88.4 $\pm$ 0.3	88.5 $\pm$ 1.6	89.3 $\pm$ 1.9	86.7 $\pm$ 2.0
	AA %	71.0 $\pm$ 4.6	77.4 $\pm$ 5.4	75.9 $\pm$ 4.6	87.1 $\pm$ 5.1
Das Mestas River	OA %	88.8 $\pm$ 0.6	88.4 $\pm$ 0.8	89.1 $\pm$ 0.5	89.3 $\pm$ 0.5
	AA %	86.5 $\pm$ 0.4	86.5 $\pm$ 1.5	87.8 $\pm$ 0.6	89.8 $\pm$ 0.3
Oitavén River	OA %	93.8 $\pm$ 0.3	93.6 $\pm$ 0.4	94.4 $\pm$ 0.3	94.1 $\pm$ 0.3
	AA %	88.7 $\pm$ 1.3	89.5 $\pm$ 0.9	90.2 $\pm$ 0.9	91.1 $\pm$ 0.4
Ulla River	OA %	98.1 $\pm$ 0.4	97.1 $\pm$ 1.2	98.2 $\pm$ 0.4	97.3 $\pm$ 1.2
	AA %	95.1 $\pm$ 1.0	91.6 $\pm$ 1.6	95.4 $\pm$ 2.0	96.4 $\pm$ 1.8
Xesta Basin	OA %	96.6 $\pm$ 0.7	96.7 $\pm$ 0.7	97.6 $\pm$ 0.1	95.9 $\pm$ 1.2
	AA %	86.8 $\pm$ 1.8	84.3 $\pm$ 2.3	89.2 $\pm$ 1.0	88.7 $\pm$ 1.6
Average	OA %	93.3 $\pm$ 3.9	93.3 $\pm$ 3.8	94.1 $\pm$ 3.7	92.9 $\pm$ 3.9
	AA %	85.3 $\pm$ 8.0	84.2 $\pm$ 9.6	86.9 $\pm$ 6.9	90.4 $\pm$ 4.2

OA: Overall Accuracy; AA: Average Accuracy.

Values: mean $\pm$ std. dev. (5 runs). Bold text indicates best results per metric and dataset.

from this analysis. Firstly, in general terms, the lowest FID and highest Precision scores are achieved when employing the balancing strategy and training the auxiliary classifier on both real and synthetic images (*Balanced Dist. / Real + Syn.*). This indicates a positive correlation between the discriminative and generative tasks: the strategy that yields the most robust classification results also tends to foster the generation of higher-fidelity images.

Conversely, higher Recall values are consistently observed when preserving the original data distribution without synthetic supervision (*Original Dist. / Real Only*). This divergence suggests that in scenarios characterized by data scarcity and severe class imbalance, forcing the generator to learn a uniform distribution increases the network’s susceptibility to mode collapse. Under these constraints, the generator tends to converge towards a limited set of prototypical modes for both minority and majority classes, rather than capturing the full diversity of the underlying manifold. This phenomenon explains why, in datasets with extreme imbalance such as Eiras Dam and Mera River, the *Balanced Dist. / Real + Syn.* configuration yields extremely low Recall values (19.8% and 34.2%, respectively), whereas the baseline configuration (*Original Dist. / Real Only*) maintains significantly higher diversity, with Recall values of 45.8% and 69.3%, respectively.

The case of Mera River is particularly illustrative of this behavior. As previously noted in Table III, this dataset ex-

hibited a significant 16.1% improvement in AA when using the balanced strategy with synthetic supervision. However, Table IV reveals that this same configuration suffers from clear failure to capture the diversity of the underlying distribution, evidenced by an almost perfect Precision of 95.4% but a low Recall of 34.2% and a high FID of 98.2. These results imply that, for the specific purpose of enhancing per-class classification performance in highly imbalanced domains, the fidelity of the generated samples appears to be a more critical factor than their diversity.

C. Comparison with State-of-the-Art Methods

Table V presents a comparative analysis of the proposed method against established baselines and state-of-the-art architectures for multispectral classification. The benchmarks include standard discriminative architectures (CNN [30], ResNet [31], ViT [32]) alongside the previously discussed GAN-based frameworks: ResBaGAN [6] and EffBaGAN (Base and Small) [7] and the diffusion-based EMViT-DDPM (ViT) utilizing EBDA with $p = 2$ [8]. The quantitative results for these reference models were retrieved directly from [7] and [8]. It should be noted that the standard deviations in the final “Average” row are calculated across the entire set of experiments for all datasets to reflect global model stability, differing from the aggregation method used in the original study. The results are reported in terms of OA and AA, with the proposed StyleGAN3 model trained using the balanced

TABLE IV
EVALUATION OF GENERATIVE MANIFOLD QUALITY UNDER DIFFERENT BALANCING AND SYNTHETIC DATA STRATEGIES

Dataset		Original Dist. Real Only	Balanced Dist. Real Only	Original Dist. Real + Syn.	Balanced Dist. Real + Syn.
Eiras Dam	FID ↓	379.1 ± 3.0	51.1 ± 2.0	71.7 ± 2.2	58.0 ± 2.5
	Precision % ↑	32.4 ± 2.9	45.3 ± 2.5	39.5 ± 2.7	53.3 ± 3.0
	Recall % ↑	45.8 ± 0.8	24.8 ± 0.6	44.5 ± 0.7	19.8 ± 0.6
Ermidas Creek	FID ↓	75.7 ± 5.7	37.2 ± 2.5	132.2 ± 5.6	20.0 ± 2.0
	Precision % ↑	67.5 ± 1.5	71.9 ± 2.3	60.5 ± 2.4	72.8 ± 2.1
	Recall % ↑	71.7 ± 0.7	47.3 ± 0.8	59.4 ± 0.8	50.6 ± 0.7
Ferreiras River	FID ↓	9.0 ± 1.0	6.0 ± 0.8	11.0 ± 0.7	12.4 ± 1.0
	Precision % ↑	84.1 ± 1.4	79.2 ± 4.1	83.5 ± 1.4	85.5 ± 3.7
	Recall % ↑	80.9 ± 0.8	61.2 ± 0.8	84.7 ± 0.8	56.4 ± 0.7
Mera River	FID ↓	45.0 ± 5.4	112.5 ± 6.4	49.5 ± 6.3	98.2 ± 7.1
	Precision % ↑	76.9 ± 2.6	95.3 ± 1.4	76.1 ± 3.2	95.4 ± 1.6
	Recall % ↑	69.3 ± 0.7	37.1 ± 1.0	77.9 ± 0.6	34.2 ± 0.7
Das Mestas River	FID ↓	8.9 ± 0.5	32.5 ± 1.3	7.8 ± 0.6	50.5 ± 1.1
	Precision % ↑	86.8 ± 1.3	78.2 ± 1.8	89.5 ± 1.5	78.3 ± 1.5
	Recall % ↑	82.8 ± 0.7	82.2 ± 0.8	85.2 ± 1.0	80.5 ± 0.8
Oitavén River	FID ↓	64.2 ± 3.0	39.5 ± 3.0	31.6 ± 3.2	27.6 ± 2.8
	Precision % ↑	47.4 ± 2.3	51.6 ± 2.1	61.2 ± 2.7	68.5 ± 1.9
	Recall % ↑	58.4 ± 0.8	44.7 ± 0.9	59.5 ± 0.8	33.5 ± 0.8
Ulla River	FID ↓	39.1 ± 1.1	99.7 ± 2.5	106.8 ± 3.1	57.5 ± 2.2
	Precision % ↑	13.8 ± 3.0	11.3 ± 2.5	14.2 ± 2.9	16.7 ± 3.5
	Recall % ↑	56.5 ± 0.8	38.5 ± 0.9	40.0 ± 1.3	42.0 ± 1.2
Xesta Basin	FID ↓	24.5 ± 1.7	82.3 ± 5.0	30.2 ± 2.8	52.9 ± 3.3
	Precision % ↑	73.7 ± 2.4	93.7 ± 1.4	80.0 ± 2.5	90.0 ± 2.8
	Recall % ↑	59.6 ± 1.0	19.1 ± 0.7	55.1 ± 1.1	20.3 ± 0.9
Average	FID ↓	80.7 ± 2.7	57.6 ± 2.9	55.1 ± 3.1	47.1 ± 2.8
	Precision % ↑	60.3 ± 2.2	65.8 ± 2.2	63.1 ± 2.4	70.1 ± 2.5
	Recall % ↑	65.6 ± 0.8	44.4 ± 0.8	63.3 ± 0.9	42.2 ± 0.8

FID: Fréchet Inception Distance. Arrows (↓ and ↑) indicate direction of improvement.
Values: mean ± std. dev. (5 runs). Bold text indicates best results per metric and dataset.

synthetic strategy that showed the best performance in the prior ablation study.

The quantitative results demonstrate that the proposed method achieves the second-highest average AA across all datasets ($90.4 \pm 4.2\%$), significantly surpassing the standard ViT baseline (87.7%) by a margin of 2.7%, and is outperformed only by the diffusion-based EMViT-DDPM framework (93.7%).

A direct comparison with the ResNet baseline provides insight into the impact of the adversarial training framework, as the discriminator of StyleGAN3 utilizes a ResNet-like backbone. The results show that the StyleGAN3 classifier consistently outperforms the standard ResNet in AA. This performance gap is particularly significant in the most challenging datasets—Mera River, Eiras Dam, and Ferreiras River—where the improvement reaches +18.5%, +10.1%, and +7.1%, respectively. As detailed in Table I, these basins contain classes with extremely limited sample sizes. These

findings confirm that integrating the ResNet architecture into the GAN’s auxiliary task significantly enhances its ability to generalize across minority classes compared to a standalone ResNet trained with standard supervision.

Excluding the EMViT-DDPM framework, the proposed method outperforms all other evaluated architectures in datasets characterized by severe data scarcity. In the Mera River, Ferreiras River and Ulla River datasets, the StyleGAN3-based classifier establishes a distinct advantage over the ViT classifier, surpassing it by substantial margins in AA of 15.5%, 5.1%, and 3.6% respectively. It is worth noting that in these extreme regimes, specialized imbalance-aware models like ResBaGAN and EffBaGAN fall below the ViT performance. In contrast, the proposed framework maintains high per-class accuracy, suggesting that the generative dynamics of StyleGAN3 offer superior regularization for minority classes when training data is critically scarce.

A distinct behavior is observed in the Das Mestas River

dataset. Here, the proposed method achieves the highest AA (89.8%), surpassing the EMViT-DDPM framework (89.2%). However, this result requires a nuanced interpretation. Unlike the other basins, Das Mestas exhibits significantly lower data scarcity, resulting in a “bounded” performance landscape where the gap between the lowest and highest performing models is remarkably narrow ($< 2.5\%$). This lack of variance suggests that when data scarcity is not the limiting factor, the performance differentiation between architectures diminishes. Consequently, while the proposed method proves effective in this regime, this slight advantage should be viewed as a specific adaptation to less critical scenarios rather than a trend generalizable to high-scarcity environments.

D. Assessment of Generative Quality and Diversity

To evaluate the fidelity and diversity of the synthesized data, we compare the proposed StyleGAN3 framework against the diffusion-based MViT-DDPM using FID, Precision, and Recall metrics. Table VI summarizes these results across all datasets.

The diffusion model (MViT-DDPM) exhibits markedly superior performance across all generative metrics, achieving the lowest FID scores and Recall values consistently exceeding 80%. This aligns with recent findings in the literature suggesting that diffusion models generally outperform GANs in modeling complex data distributions [17]. In contrast, the StyleGAN3 model yields significantly higher FID scores and lower Recall values. For instance, in the Eiras Dam and Xesta Basin datasets, Recall drops to approximately 20%, indicating that the GAN captures a much narrower subset of the data distribution compared to the diffusion model.

However, an analysis of the Precision metric reveals a more nuanced landscape. While the StyleGAN3 does not match the diffusion model, it maintains respectable Precision levels—often exceeding 70% and reaching up to 95.4% in the Mera River dataset. This suggests that while the diversity of the generated samples is limited, the samples that are produced possess a reasonable degree of realism and adherence to the training manifold.

This disparity in performance, particularly in Recall, should be interpreted within the context of the distinct training objectives of each architecture. The MViT-DDPM is optimized strictly for high-fidelity image synthesis. Conversely, the proposed StyleGAN3 framework is constrained by the auxiliary classifier, which likely guides the generator toward a subspace of features that are most discriminative for the classification task. We hypothesize that this “low-diversity” generation is sufficient to populate the essential gaps in the class manifolds—providing the necessary support for the classifier in data-scarce scenarios—without requiring the exhaustive distribution coverage achieved by diffusion models.

The qualitative assessment presented in Fig. 5 corroborates these quantitative findings. The samples generated by MViT-DDPM, particularly for the Bare soil, Meadows, and Native trees classes, exhibit a high degree of realism, rendering them nearly indistinguishable from the real patches shown on the right. The StyleGAN3 generated images also display realistic

textures; however, a visual inspection reveals a recurrence of specific patterns and orientations. This effect is most notable in the Bare soil and Native trees classes, where the limited diversity manifests as repetitive textural motifs. While visually less varied, these patterns seemingly retain the critical spectral-spatial structures required to enhance the robustness of the auxiliary classifier.

Finally, strictly regarding computational efficiency, the proposed framework demonstrates a clear advantage over diffusion-based alternatives. While the EMViT-DDPM requires approximately 48 hours for convergence on the largest datasets [8], our StyleGAN3 adaptation significantly reduces this computational burden. Table VII details the average training times recorded across all experimental configurations shown in Table III. As observed, the model consistently converges in approximately 9 hours 30 minutes for the largest dataset, representing more than a 4x speedup compared to the state-of-the-art diffusion baseline.

VI. DISCUSSION

This study addresses the challenge of pixel-level classification in high-resolution multispectral imagery characterized by severe data scarcity and class imbalance. To this end, we proposed a modification of the StyleGAN3 architecture [20]. By integrating an auxiliary classification head and employing a class-balanced training strategy, we successfully mitigated the effects of severe class imbalance. This approach proves particularly valuable not only in the context of remote sensing but also in broader computer vision domains where datasets frequently exhibit long-tailed distributions and limited ground truth availability.

Building upon recent state-of-the-art benchmarks established by ResBaGAN [6] and EffBaGAN [7], our results validate the effectiveness of the proposed architecture. As evidenced in Table V, the integration of the auxiliary classifier into the StyleGAN3 backbone consistently outperforms these previous GAN-based frameworks in terms of Average Accuracy (AA). This improvement is most pronounced in datasets with critical scarcity, such as Mera River and Ferreiras River, where our method surpasses standard ResNet and ViT baselines by significant margins. This suggests that the nature of StyleGAN3, combined with active synthetic supervision, provides a robust feature extraction foundation for the discriminator.

However, a comparison with the diffusion-based EMViT-DDPM framework [8] highlights the distinct trade-offs between generative approaches. Theoretical foundations [17] posit that diffusion models often generate superior synthetic samples compared to GANs, albeit at a higher computational cost. Our findings reflect this distinction: the EMViT approach achieves higher image quality metrics—such as FID and Recall in Table VI—which contribute to its high classification performance.

Yet, this performance gap must be weighed against computational efficiency. Whereas MViT training necessitated approximately 48 hours for the largest datasets [8]—specifically

TABLE V
COMPARATIVE ANALYSIS WITH STATE-OF-THE-ART ARCHITECTURES

Dataset		CNN	ResNet	ViT	ResBaGAN	EffBaGAN Base	EffBaGAN Small	EMViT-DDPM ViT	StyleGAN3 (ours)
Eiras Dam	OA %	94.4 ± 1.5	89.8 ± 7.0	97.7 ± 0.1	97.2 ± 1.5	97.9 ± 0.8	98.1 ± 0.5	95.8 ± 0.1	93.9 ± 1.2
	AA %	71.8 ± 3.3	79.9 ± 7.3	89.4 ± 0.8	84.9 ± 2.8	85.9 ± 2.2	90.7 ± 1.5	94.3 ± 0.3	90.0 ± 1.8
Ermidas Creek	OA %	96.0 ± 1.4	98.1 ± 0.7	98.8 ± 0.1	98.6 ± 0.5	98.6 ± 0.3	98.8 ± 0.2	97.8 ± 0.1	96.8 ± 0.6
	AA %	89.3 ± 2.7	94.8 ± 1.6	95.6 ± 0.0	95.8 ± 1.0	94.6 ± 2.3	96.2 ± 0.4	97.3 ± 0.2	96.1 ± 0.4
Ferreiras River	OA %	91.9 ± 0.2	92.8 ± 0.3	92.4 ± 0.1	92.6 ± 0.3	93.1 ± 0.2	93.0 ± 0.1	91.1 ± 0.1	89.5 ± 1.3
	AA %	74.7 ± 1.0	76.8 ± 0.9	78.8 ± 0.1	75.8 ± 1.1	77.4 ± 0.2	77.1 ± 0.4	87.0 ± 1.7	83.9 ± 5.0
Mera River	OA %	92.5 ± 0.2	92.8 ± 0.2	92.8 ± 0.1	92.6 ± 0.7	93.3 ± 0.2	92.9 ± 0.4	89.0 ± 0.1	86.7 ± 2.0
	AA %	68.3 ± 1.4	68.6 ± 1.9	71.6 ± 3.9	70.5 ± 4.9	68.4 ± 1.2	67.2 ± 0.8	91.8 ± 0.2	87.1 ± 5.1
Das Mestas River	OA %	91.7 ± 0.3	92.1 ± 0.3	91.2 ± 0.1	92.0 ± 0.2	92.3 ± 0.2	91.9 ± 0.6	90.0 ± 0.1	89.3 ± 0.5
	AA %	87.5 ± 0.9	88.8 ± 0.9	87.8 ± 0.2	87.8 ± 1.2	89.2 ± 0.7	88.9 ± 0.7	89.2 ± 0.1	89.8 ± 0.3
Oitavén River	OA %	95.2 ± 0.3	96.2 ± 0.5	97.0 ± 0.2	97.2 ± 0.3	97.5 ± 0.1	97.4 ± 0.2	95.2 ± 0.1	94.1 ± 0.3
	AA %	88.4 ± 1.1	90.6 ± 1.2	92.3 ± 0.6	92.3 ± 1.6	92.8 ± 1.0	92.2 ± 0.9	95.0 ± 0.2	91.1 ± 0.4
Ulla River	OA %	97.0 ± 1.5	98.3 ± 1.7	97.9 ± 0.8	96.2 ± 0.5	89.3 ± 16.0	98.4 ± 0.9	97.5 ± 0.2	97.3 ± 1.2
	AA %	91.0 ± 5.6	95.3 ± 4.0	92.8 ± 4.2	84.5 ± 9.1	86.6 ± 11.3	93.1 ± 3.2	98.7 ± 0.2	96.4 ± 1.8
Xesta Basin	OA %	97.2 ± 1.2	98.4 ± 0.6	98.6 ± 0.0	98.5 ± 0.2	99.2 ± 0.1	98.8 ± 0.2	98.0 ± 0.1	95.9 ± 1.2
	AA %	88.8 ± 2.8	92.2 ± 1.6	93.1 ± 0.5	86.2 ± 5.3	93.9 ± 0.4	91.9 ± 2.5	96.7 ± 0.1	88.7 ± 1.6
Average	OA %	94.5 ± 2.2	94.8 ± 3.3	95.8 ± 3.1	95.6 ± 2.8	95.2 ± 3.6	96.2 ± 3.0	94.3 ± 3.7	92.9 ± 3.9
	AA %	82.5 ± 9.2	85.9 ± 9.7	87.7 ± 8.3	84.7 ± 8.2	86.1 ± 9.1	87.1 ± 9.9	93.7 ± 4.1	90.4 ± 4.2

OA: Overall Accuracy; AA: Average Accuracy.

Values: mean ± std. dev. (5 runs). Background colors denote 1st, 2nd, and 3rd best results.

Xesta Basin—the StyleGAN3 architecture demonstrates superior efficiency, converging in significantly less time (around 9 hours 30 minutes). Consequently, further work is required to determine if we can bridge this classification gap using the faster StyleGAN3 architecture purely as a generator. Future research should explore training StyleGAN3 without the auxiliary head and incorporating it into the Equilibrium-Based Data Augmentation (EBDA) algorithm proposed by [8]. This would enable a precise evaluation of the trade-off between the rapid inference capabilities of GANs and the generative quality of diffusion models, offering a clear perspective on the balance between computational resources and classification accuracy.

This balance is further evidenced by the analysis of generative metrics in Table IV. Experimental results indicate that the configuration yielding the highest AA (*Balanced Dist. / Real + Syn*) often exhibits high Precision but low Recall. This trend is particularly evident in the highly imbalanced Mera River and Ferreiras River datasets, where fidelity appears to be more relevant than diversity for improving classification results. These results raise a fundamental question for future investigation: does ensuring that generated samples fall strictly within the real data manifold (high fidelity) matter more for data augmentation than generating diverse samples that might drift from the distribution? The answer to this question could significantly influence the choice of generative architectures in

resource-constrained environments.

Finally, while the proposed StyleGAN3 modification is specifically demonstrated on high-resolution multispectral imagery captured by UAVs, the framework possesses inherent adaptability to satellite imagery with varying spatial resolutions. The core generative and classification components can be adjusted for different resolution levels. However, the pre-processing stage, specifically superpixel segmentation, would require careful tuning based on spatial resolution.

VII. CONCLUSION

In this work, a unified framework based on the StyleGAN3 architecture is proposed for high-resolution multispectral image classification. Originally designed for high-fidelity photorealistic synthesis, StyleGAN3 distinguishes itself through an alias-free design that treats images as continuous signals to ensure geometric consistency. Our proposal addresses the challenges of data scarcity and class imbalance—typical of drone-based multispectral remote sensing—by integrating an auxiliary classification head into the discriminator and employing a balanced training strategy that exploits the available spectral bands. Crucially, the combination of this generative backbone with superpixel-based patch extraction ensures the preservation of critical spectral and spatial structures. Finally, to mitigate imbalance, we introduce a uniform sampling strat-

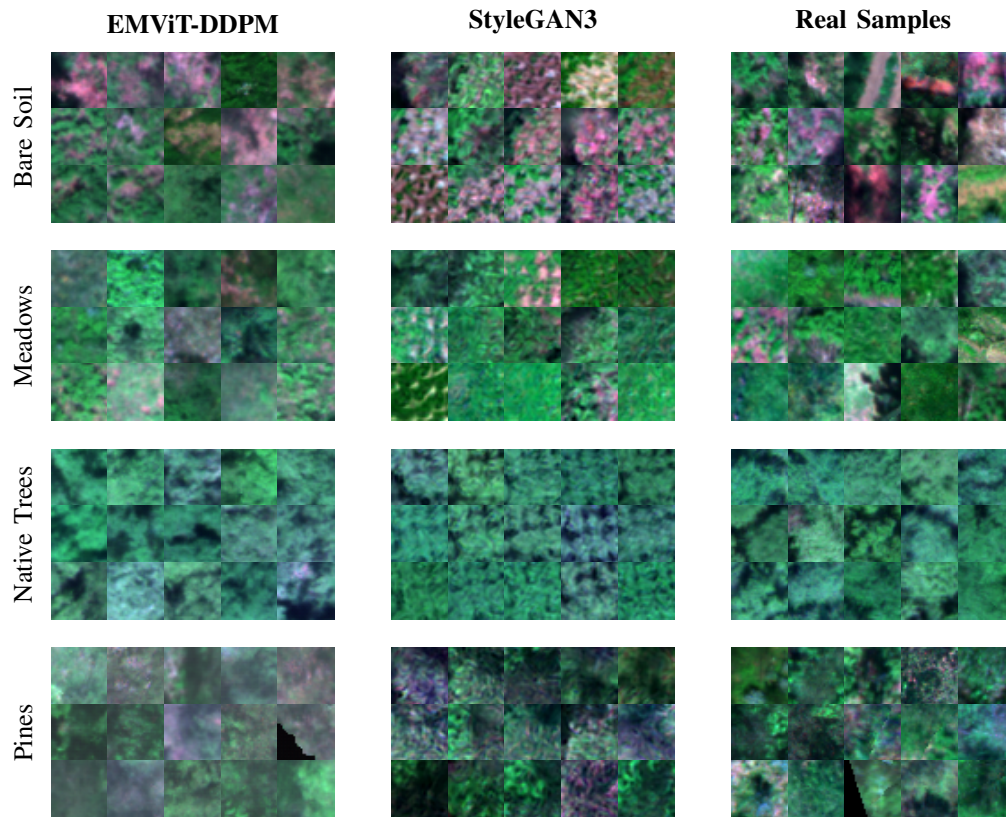


Fig. 5. Qualitative comparison of synthetic samples generated by EMViT-DDPM (left column) and StyleGAN3 (middle column) against real samples (right column) from the minority classes of the Das Mestas River dataset. Each row corresponds to a different class: Bare Soil, Meadows, Native Trees, and Pines. The visual fidelity and diversity of the generated samples are assessed in relation to the real data.

egy during training that forces the generator to produce balanced samples, providing robust and regularized supervision for the auxiliary classifier.

To effectively evaluate the generated samples and their impact, we adopted an evaluation methodology combining fidelity (Precision), diversity (Recall), and an adapted FID metric. Additionally, the use of a judge model specifically trained for remote sensing data ensures comprehensive and domain-relevant assessments of synthetic sample quality.

The experiments reveal an interesting trade-off between fidelity and diversity. Although our balanced training strategy induced mode collapse in minority classes (low Recall), the generated samples maintained high fidelity (high Precision), which proved decisive for improving the auxiliary classifier's performance. This suggests that, in the context of GAN architectures with integrated classifiers, visual precision and proximity to the real manifold are more critical than exhaustive distribution coverage for regularization purposes.

The results, obtained on eight high-resolution multispectral datasets captured by UAVs, demonstrate the effectiveness of the proposed framework. In particular, our StyleGAN3 architecture achieved a mean Average Accuracy (AA) of 90.4%, significantly outperforming baseline discriminative models (such as ViT and ResNet) and previous specialized GAN architectures (ResBaGAN and EffBaGAN) in critical scarcity

scenarios. The framework consistently reduced classification bias for underrepresented classes, highlighting its ability to address the challenges of both data scarcity and class imbalance.

Future work should focus on decoupling the generative component from the classification task to perform a more extensive comparison with state-of-the-art data augmentation methods. Specifically, training StyleGAN3 as a standalone generator within standard balanced data augmentation protocols would allow for a direct analysis of the cost-benefit relationship between the superior computational efficiency of GANs and the synthesis characteristics of diffusion models. This would provide valuable insights for deploying generative augmentation in scenarios with strict time or computational constraints.

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TABLE VI
GENERATIVE QUALITY AND DIVERSITY COMPARISON BETWEEN
MVIT-DDPM AND STYLEGAN3

Dataset		MViT-DDPM	StyleGAN3
Eiras Dam	FID ↓	0.9 ± 0.0	58.0 ± 2.5
	Precision % ↑	80.6 ± 0.4	53.3 ± 3.0
	Recall % ↑	88.1 ± 0.5	19.8 ± 0.6
Ermidas Creek	FID ↓	0.7 ± 0.0	20.0 ± 2.0
	Precision % ↑	87.6 ± 0.6	72.8 ± 2.1
	Recall % ↑	90.1 ± 0.2	50.6 ± 0.7
Ferreiras River	FID ↓	1.9 ± 0.1	12.4 ± 1.0
	Precision % ↑	91.5 ± 1.7	85.5 ± 3.7
	Recall % ↑	83.2 ± 0.4	56.4 ± 0.7
Mera River	FID ↓	4.7 ± 0.1	98.2 ± 7.1
	Precision % ↑	86.7 ± 0.4	95.4 ± 1.6
	Recall % ↑	90.9 ± 0.7	34.2 ± 0.7
Das Mestas River	FID ↓	1.5 ± 0.3	50.5 ± 1.1
	Precision % ↑	93.7 ± 0.3	78.3 ± 1.5
	Recall % ↑	90.1 ± 1.1	80.5 ± 0.8
Oitavén River	FID ↓	2.1 ± 0.1	27.6 ± 2.8
	Precision % ↑	84.6 ± 0.5	68.5 ± 1.9
	Recall % ↑	87.1 ± 0.3	33.5 ± 0.8
Ulla River	FID ↓	16.5 ± 0.3	57.5 ± 2.2
	Precision % ↑	63.2 ± 1.9	16.7 ± 3.5
	Recall % ↑	79.1 ± 0.3	42.0 ± 1.2
Xesta Basin	FID ↓	2.1 ± 0.1	52.9 ± 3.3
	Precision % ↑	85.6 ± 0.6	90.0 ± 2.8
	Recall % ↑	89.7 ± 0.5	20.3 ± 0.9

FID: Fréchet Inception Distance. Arrows (↓ and ↑) indicate direction of improvement. Values: mean ± std. dev. (5 runs).

TABLE VII
TRAINING TIME PER DATASET

Dataset	Training Samples	epochs	Training Time (hours)
Eiras Dam	2309	400	2.36 ± 0.00
Ermidas Creek	3474	400	3.56 ± 0.01
Ferreiras River	8180	400	8.42 ± 0.04
Mera River	8992	400	9.35 ± 0.06
Das Mestas River	5877	400	6.06 ± 0.02
Oitavén River	3240	400	3.33 ± 0.02
Ulla River	217	800	0.45 ± 0.00
Xesta Basin	9206	400	9.51 ± 0.05

Training times were recorded using 2 NVIDIA A100 (40 GB). Values represent the mean ± std. dev. over 20 runs per dataset (4 conf. × 5 runs).

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